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Photobiomodulation for pain relief: Model-based estimates of effective doses of light at the neural target

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Abstract

Introduction: Photobiomodulation (PBM) has been studied since the 1960s as a clinical tool. More recently, PBM has been observed to reduce compound action potential components and hypersensitivities associated with neuropathic pains. However, no definitive description of efficacious light parameters has been determined. Some reasons may be that previous meta-analyses and reviews have focused on emitter output rather than the light at the target tissue and have included data sets that are large but with notable variability (e.g., combining data from various disease etiologies, and data from PBM at various wavelengths). This fact has made it difficult to successfully define the range of effective parameters.

Methods: In this study, photon propagation software was used to estimate irradiance at a target nerve using several published data sets chosen for their narrow criteria to minimize variability. Utilizing these estimates, effective and ineffective light irradiances at the nerve of interest for wavelengths of 633 nm or 808–830 nm were examined and estimated. These estimates are focused on the amount of light required to achieve a reduction in pain or a surrogate measure via a hypothesized nerve block mechanism.

Results: Accounting for irradiance at the target nerve yielded a clear separation of PBM doses that achieved small-fiber nerve block from those that did not. For both the 633 nm group and the 808–830 group, the irradiance separation threshold followed a nonlinear path with respect to PBM

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Andrew Buzza: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kalista Tapas:** Writing – original draft, Methodology, Investigation. **Juanita Anders:** Writing – review & editing, Funding acquisition. **Michael Jenkins:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Michael Moffitt:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Formal analysis, Conceptualization.

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application duration, where shorter durations required higher irradiances, and longer durations required lower irradiances. Using the same modeling methods, irradiance was estimated as a function of depth from a transcutaneous source (distance from skin surface) for emitter output power using small or large emitter sizes.

Conclusion: Taken together, the results of this study can be used to estimate effective PBM dosing schemes to achieve small-fiber inhibition for various anatomical scenarios.

Keywords

Photobiomodulation; Dosimetry; Pain relief; Peripheral nerve; Photon propagation

1. Introduction

1.1. Photobiomodulation: Complex Multifaceted Treatment Opportunity

Groundbreaking work from Mester et al. in 1968 [1] demonstrated the surprising capability of red light to induce hair growth in mice. Subsequent studies have shown that red, near infrared (NIR), and infrared light induce many favorable biological effects that could be of great therapeutic utility including reducing inflammation [2,3], promoting wound healing [4,5], mitigating oxidative stress [6], promoting cell survival [7], and at higher irradiances – selective small-fiber block [8–11]. Thus, the use of light at therapeutic doses holds promise for the treatment of many different neurological pathologies such as stroke [12], traumatic brain injury [12], dementia [13], Parkinson's disease [14,15], ophthalmic neurodegeneration [16], and pain syndromes [17,18] including peripheral neuropathies, fibromyalgia, and others. The use of light for these purposes has become a field of study commonly referred to as photobiomodulation (PBM) with the therapeutic application of this phenomenon termed photobiomodulation therapy (PBMT).

However, the ability to achieve effective light dosing of target tissues and the mechanisms of the subsequent induced effects are complex processes that are not completely understood. Although numerous candidate photo-acceptors have been identified from cytochrome C oxidase of the respiratory chain [19], to light-sensitive ion channels [20], to water molecules (higher wavelengths) [20]. Specific mechanisms of action for particular effects have not been completely determined and may be due to complementary roles between multiple photo-acceptors.

Additionally, because tissue optical properties differ across wavelengths, the penetration depth of light transmission varies according to wavelength and tissue type. Reports of clinical and pre-clinical PBMT describe various types and thicknesses of tissue between the light source and the target tissue [21–24], decreasing the amount of light reaching the target. Finally, beyond wavelength, the PBM dose encompasses a wide range of parameters such as irradiance, duration of application, continuous wave vs. pulsatile delivery, etc., resulting in an extensive parameter space.

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1.2. Efforts to Understand PBM Dose

Predicting effective PBMT dosing protocols has been very difficult because of the complexities related to: incomplete understanding of mechanisms, light delivery, and the large parameter ranges used. Several groups have provided insights into effective PBM dosing, often through meta-analysis of multiple studies [18,22,24,25]. These meta-analyses have typically included data with a broad range of outcomes, cell types, wavelengths, dosing mechanisms, and outcome types (ranging from wound healing to action potential block). Although these analyses include large amounts of data, very broad inclusion criteria can confound the ability to develop strong conclusions.

Additionally, the vast majority of PBMT work has included transcutaneous light application, such that light must traverse intervening tissue layers of various compositions, thicknesses, and optical properties to reach the target tissue. The amount of light that arrives at the target tissue is difficult to measure and is not typically provided in the literature. To our knowledge, all previous meta-analyses have described the light parameters in terms of emitter output. This approach has enabled authors to draw broad conclusions, such as the observation that the success rate of PBMT for pain reduction increases with increased emitter power and emission duration. But, the authors appropriately note that the broad parameter range makes it difficult to draw more specific conclusions. Further, the description of trends as well as guidelines describing transcutaneous applications (like those presented by the World Association of Laser Therapy, WALT) do not apply for near-nerve treatment paradigms.

1.3. A Unique Model-Based Approach to Evaluate Dosing (Focused on PBM for Pain Via Nerve Block)

As an alternative approach, we analyzed studies in the literature to derive predictions of effective PBMT dosing. First, narrow acceptance criteria were determined to select studies for evaluation that used a common wavelength and that were plausibly operating by the same PBM mechanism, rather than including larger amounts of data with the accompanying variability. Second, those studies were analyzed using photon propagation models to enable the comparison of the irradiances of light at the target nerve tissue for each study, rather than the typical approach of comparing light properties at the emitter.

In particular, PBMT studies related to the treatment of pain via a hypothesized nerve block mechanism using either 633 or 808–830 nm light were analyzed. PBM to induce a selective small-fiber nerve block has useful properties for evaluating our approach, including a specific anatomical target (namely, the nerve that is to be inhibited), quantifiable behavioral and/or electrophysiological effects, and translational interest.

Although optimal dosing of PBMT to achieve this effect has been unclear, several groups have observed reductions in pain and hypersensitivities with NIR light application, 808 to 830 nm, in both humans and animal models [11,26–28]. At 808 to 830 nm, using varying nerve preparations, researchers have also reported increases in membrane potential [29], decrease in small-fiber action potential amplitude [8–10], and overall suppression of action potentials [30]. Additionally, at wavelengths near 633 nm, a reduction of action potential

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(AP) amplitude in ex vivo nerve preparations [31–33] has been observed. Rather than compare data based on the emitter output, we hypothesized that simple geometric models could be used to provide better estimates of the light at the nerve and enable comparison with less “noise” associated with anatomical differences. We termed this “normalization” method PHOtons at the Target for Outcome Standardization (PhoTOS).

The goals of the present study were to: (1) highlight the value of using model-based light estimates at the target tissue to understand PBMT dosing; (2) provide estimates of the light-at-nerve ranges where a level of neural inhibition using 633 nm or using 808–830 nm light was achieved; and (3) provide insight to guide decision-making for transcutaneous PBMT to achieve small fiber inhibition.

2. Methods

2.1. Literature Review and Inclusion / Exclusion Criteria

To estimate the amount of light required to cause the biological effect of interest, geometric models of light propagation were generated for several experimental scenarios reported in the literature.

2.1.1. PBM Indication - Neural Inhibition and Pain Relief—We sought to determine the irradiance and duration combination (PBMT dose) *at the nerve* necessary to reduce pain or pain symptoms, or to inhibit action potential activity when administering laser treatments in continuous wave mode at the wavelengths described below. Both evaluations of light applications on in vivo and ex vivo nerves were included, as long as other criteria were met. Two wavelength ranges were included and analyzed independently – 633 nm, and 808–830 nm. These two wavelengths were chosen because they are the most frequently used wavelengths in evaluations of PBMT for pain in the literature [23,24].

Experiments that measured patient reported pain, hypersensitivities in animal models, and relevant electrophysiological effects were included. Action potential amplitude, latency, and compound action potential decreases were included because they correlate with a reduction of neuropathic and nociceptive pain [34,35].

2.1.2. Number of PBMT Applications—Only experiments using a single application of PBMT were included for additional analysis in this study. This inclusion criterion was prompted by the desire to observe only the initial effects of PBMT related to pain relief and to manage complexity. For example, although multiple applications of PBMT may increase effect size over time, these clinically reasonable dosing schemes were not used in this first study using model-based methods to estimate effective PBM dose.

2.1.3. Organism Class—Only experiments from the class Mammalia were included, since axonal membrane properties are known to differ between animal classes [36,37] and effects of light at other wavelengths in different classes have been shown to have notably different electrophysiological responses [38].

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2.1.4. Exclusion Criteria—Exclusion criteria were the following: experiments with central nervous system targets, non-mammalian nerves, pulsatile light delivery paradigms, and non-pain in vivo and in vitro models.

Consistent with the inclusion and exclusion criteria described above, candidate studies were identified using Google Scholar and Pubmed (January 1968 to May 2022). The search terms were “LLLT”, “Photobiomodulation”, “808”, “633”, “830”, “632”, “810”, “Pain”, “Action Potential”, “Electrophysiology”, “Irradiation”, “Laser” and permutations of these terms. References of previous systematic reviews and meta-analyses were also searched with the same criteria. After application of the inclusion and exclusion criteria described above, experiments from 8 publications using 808 to 830 nm (Table 1) and 5 publications using 633 nm (Table 2) were included for evaluation using our model-based approach (Fig. 1).

2.2. Definition of “Effective”

Effective is defined as a result that is statistically significantly changed compared to the untreated state, observed after a single dose within a 24-h post treatment time frame. Of note, although not different from the baseline state, post hoc analysis of the experiment of Sato et al. [10] was found to be statistically significant after 60 s of laser treatment when compared to the pain state of the group’s pain experiment from the same study, and therefore this experiment was included in the present study.

2.3. PHOtons at the Target for Outcome Standardization (PhoTOS)

The combination of selecting a specific disease state and use of photon propagation estimates has yet to be performed for the purposes of normalizing PBM dosing for pain. Herein, this approach is referred to as PHOtons at the Target for Outcome Standardization, or PhoTOS.

2.3.1. Modeling Photon Propagation—To develop biologically relevant approximations of photon propagation in the body, we used MMCLab (version 2019) [42,43], a Matlab™-based software package published by the Fang group. In MMCLab, relevant geometries are modeled with a mesh composed of tetrahedral elements using COMSOL™ (Fig. 2A). The optical parameters of the various tissue types were based on values from the literature (Table 3). For each experiment that was modeled, a simple rectangular prism was constructed with various tissue domains to mimic the experimental set up. Tissue thicknesses were estimated based on data from the literature (Table 4). The mesh size was chosen to maintain at least five hundred thousand tetrahedra in each model. Monte Carlo estimates of light propagation and absorption were generated by injecting 50 million photons into each experimental model. By way of example, for Tsuchiya et al., 1993, a nerve was surrounded by air above and with a layer of muscle underneath the nerve [9] (Fig. 2C).

2.3.2. Simplifying Assumptions and Model Sensitivity Analysis—Due to the lack of detailed anatomical information from each experiment, assumptions needed to be made. Model assumptions included: the emitter has a divergence angle of 0°; the emitter irradiance is uniform across the emitter surface; the tissue properties are uniform within a

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tissue type and do not change as a result of the irradiation, no blood vessels were modeled, and the nerve is cylindrical. Sensitivity of these assumptions was not specifically evaluated, but a sensitivity analysis of tissue optical parameters, which are known to vary, was performed as well as varying the divergence angle to that of an optical fiber with a numerical aperture (NA) of 0.39 (22.95°). To evaluate the impact of changes of optical parameters on the results and conclusions, we evaluated whether conclusions changed for models where tissue optical properties (scattering and absorption coefficients) were modified by ±20%. Anisotropy typically varies by less than 5%, so these values were left unchanged [41–43].

2.3.3. Model Output—The MMCLab output includes the irradiance as a function of space throughout the model geometry Fig. 2b). In this study, the maximum irradiance at the target nerve was used for our PhoTOS normalization.

For each experiment, the maximum irradiance at the nerve and light-application time were plotted, and parameters that had been statistically deemed “effective” or “not effective” were displayed. To highlight the difference from PhoTOS, a similar plot was developed using emitter characteristics – the standard approach.

2.4. Light Penetration Models for General Reference and Validation in Cadaveric Tissue

Given that PBMT is most often delivered transcutaneously, using the same optical parameters that were used to derive the PhoTOS-based estimates, models of light penetration through skin and into soft tissue (muscle) were developed, and can be used for general reference to estimate the effectiveness of a PBMT dose for selective small fiber block. The reference models consisted of a light source directly on a section of skin (2 mm thick) with soft tissue (muscle) beneath it. The light penetration (maximum irradiance) was modeled, plotted, and tabulated as a lookup table for reference. The nerve was not explicitly modeled because it was assumed that the presence of the nerve would have only a modest effect on the estimate of light at the nerve surface. The following permutations were modeled: (1) a 1 mm diameter light source representative of PBMT using a laser and optical fiber (as this is a common optical fiber diameter); (2) a 25 mm diameter light source representative of PBMT using a light emitting diode-based delivery system (similar size to FDA cleared commercial products from various vendors).

The development of a simple reference model also afforded the opportunity to validate the general modeling approach by comparing the output of the reference model with measurements of light transmission through excised rat tissue. Sections of muscle (biceps femoris) and skin were removed from a rat to validate the modeling approach. An 808 nm wavelength laser with an output power of 1 W (1 mm diameter optical fiber) was applied directly to a tissue slab that included a layer of skin (2 mm thick), and up to three additional layers of muscle (2 mm per layer) beneath the skin. Each layer of tissue had a cross sectional area of approximately 1 cm^2. A power sensor (PowerMax, Coherent Scientific, Santa Clara, CA) with a cross-sectional area of 2.27 cm^2 was used to measure the power of the light exiting the tissue, and then calculate the average power density. This irradiance value was compared to a model of the same scenario.

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3. Results

3.1. Effective and Ineffective PBMT Doses – Comparison of Emitter Properties and PhoTOS

Effective and ineffective PBMT doses were plotted (irradiance vs. application time) for 808–830 and 633 nm based on irradiance at the emitter (Fig. 3A, Fig. 4A; standard approach) and irradiance at the target nerve using PhoTOS (Fig. 3B, Fig. 4B). Both the standard and the PhoTOS methodologies demonstrated the intuitive trend that greater irradiance and application time increases effectiveness of neural inhibition. However, the delineation between effective and ineffective parameters is particularly evident when the PhoTOS methodology is used to estimate irradiance at the target. Note that PhoTOS affects relative relationships of data points. For example, the maximum 808–830 nm PBMT doses for references 5 and 6 in Table 1 appear comparable when plotted by irradiance at the emitter (Fig. 3A), but PhoTOS reveals that the irradiance at the nerve is likely to be different by multiple orders of magnitude (Fig. 3B).

3.2. Sensitivity of PhoTOS to Model Parameters

Modeling of beams with a specific numerical aperture (NA = 0.39) was conducted to evaluate the sensitivity of the assumption of non-diverging beams (Figs. 3C and 4C). These models displayed similar effective and ineffective regions as observed without divergence. Similarly, when using modified absorption and scattering coefficients in the models, the efficacious regions separated from non-efficacious regions as previously demonstrated (Fig. 5). However, the specific estimates of effectiveness thresholds vary based on the scattering and absorption values (with percent differences ranging from 2 to 30% of the original irradiance values).

3.3. Reference Model and Comparison with Measurements in Excised Tissue

The reference model included transcutaneous delivery of 1 W of light (808 nm) into a layer of skin (2 mm) covering muscle, using either of two emitter sizes (1 mm or 25 mm in diameter). As expected, the simulations showed that a smaller emitter diameter delivered the light into a more focal region, and along the optical axis the light penetrated deeper into the tissue (Fig. 6). Delivered from the larger diameter emitter, the light did not penetrate as deep, but there was a wider region (about 10 mm width, +/− 5 mm from center) of higher and more uniform fluence beyond a depth of 2 mm.

Outputs of the reference model were compared with measurements using excised rat tissue. In the excised tissue experiment, the 1 mm diameter optical fiber applied 1 W of 808 nm light. The average power exiting the tissue was measured with a power sensor with a 2.27 cm^2 circular detection region. The experimental power density was calculated to be 0.001207 +/− 5.35×10^{-5} W/cm^2, 0.000855 +/− 6.41×10^{-5} W/cm^2, 0.000593 +/− 3.54×10^{-5} W/cm^2, and 0.000383 +/− 2.74×10^{-5} W/cm^2, for increasing numbers of layers (skin +0, 1, 2, or 3 layers of muscle) (Fig. 7). When the experiment was modeled (same modeling methods as used by the PhoTOS technique), the simulated average output power density was 0.0015 W/cm^2, 0.000779 W/cm^2, 0.000556 W/cm^2, and 0.000384 W/cm^2 at depths of 2, 4, 6, and 8 mm, respectively. The model outputs and experimental outputs trend very similarly

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(Fig. 7), and the concordance supports the use of simple models for making estimates as proposed with PhoTOS.

4. Discussion

4.1. PhoTOS – A Proposed Approach to Estimate Effective PBMT Dosing Across Studies

Dosing parameters for effective PBMT application vary widely as a function of wavelength, frequency of light application, and disease state. To determine effective dosing parameters, previous groups have sought to map effectiveness by emitter output (e.g., emitter irradiance) and fluence. As a high-level trend, they found that higher fluence and irradiance from the emitter corresponded to greater pain relief, but there were numerous exceptions for specific experimental and anatomical scenarios. We considered that this variability is at least in part attributed to differences in the irradiance/fluence of light reaching the nerves of interest, since nerves are located at different tissue depths for each type of application. For example, Tsuchiya et al. emitted photons directly onto the nerve [9], whereas Bartlett et al. delivered the photons through skin and muscle to reach the nerve [39]. Thus, we hypothesized that light irradiance and fluence reaching the target tissue could enable a better prediction of the PBMT response. Using PhoTOS, irradiance was estimated at the target tissue for the papers highlighted in Tables 1 and 2. PhoTOS estimates showed a clearer separation between the successful and unsuccessful treatment paradigms when plotted as a function of irradiance versus light application duration (Figs. 3 and 4), providing insights into effective doses for small fiber inhibition.

4.2. Applying PhoTOS to Direct PBMT for Pain

The data from a PhoTOS analysis can inform selection of dosing parameters for various purposes. Fig. 8 shows an example of how PhoTOS information supported selection of dosing parameters in a study applying direct PBMT at the nerve in a rodent model of neuropathic pain. Rough guidelines for low-level transcutaneous PBMT dosing exist, such as those presented by WALT. To our knowledge, guidelines are not available on higher doses of PBMT to achieve selective nerve block using either direct or transcutaneous PBMT for nerves of various depths. The estimates from the present PhoTOS analysis can be used to support the development of such guidelines which ultimately should also include thermal considerations (out of scope in the present analysis). The data from the reference model (Fig. 6) could be compared to the PhoTOS analysis data (Fig. 3) to estimate effective combinations of emitter power and PBMT application duration. This comparison was done speifically for the PhoTOS-transformed doses of the aforementioned study [11] (Fig. 8). The values from the transformed doses, in combination with the reference model (Fig. 6), were used to create PhoTOS-based estimates of effective parameters for various dosing strategies and distances to the target nerve (Table 5). These may be useful for future investigations, and it is appreciated that these are coarse estimates, and that the strength of the response could be further augmented by applying longer durations or higher irradiances [11].

4.3. Limitations and Next Steps

The present analysis was limited by the number of experiments that met the inclusion criteria, and by the amount of information provided on the laser parameters used. Because

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having sufficient information to generate a simplified model is critical to the PhoTOS method, many early articles that did not include important properties of the laser system, such as emitter area or diameter, could not be included. Future articles associated with the PBMT field should include the following minimal set of descriptive information: emitter output power and diameter; treatment time; wavelength; output divergence angle (or associated emitter); and the target tissue. Although the PhoTOS method did result in a separation of the dosing parameters associated with an effect and dosing parameters that did not yield an effect, more data on dosing parameters would clarify a definitive threshold for irradiance and duration of irradiation. Our analysis suggested that lower irradiances may be needed for 633 nm light compared to 808–830 nm light. However, the sparsity of data did not permit strong conclusions to be drawn because equivalent at-the-nerve dosing parameters for the two wavelengths were not available. Also, the information at the lower wavelength did not include in vivo behavioral data, making direct comparison difficult.

This analysis was limited to outputs of simplified models. There are resolution and homogeneity limits to models readily achievable in MMCLab. For example, exact emitter divergence angles could cause variation to the current results. Some of these limitations cannot reasonably be overcome, and are inherent in the approach, while others could be addressed by including more information (e.g., divergence angles) in experimental reports.

As a starting point for PhoTOS analysis, we chose the maximum irradiance at any node within the nerve domain of the model as our indicator of irradiance at the nerve. This value is readily extractable for all types of experimental paradigms. However, various other model outputs could be considered such as energy absorbed by the nerve or average irradiance across a particular length of the nerve. These alternative metrics are more sophisticated and more complex to extract from a model, and may even require larger models in some cases, but should be considered in future evaluations.

The present analysis mapped outcomes in binary: effective or not effective. However, many PBMT paradigms for pain reduction yield a graded effect that varies continuously with dose. This has been observed in the context of both electrophysiological and behavioral outcomes [9,11]. As is expected, higher doses elicit stronger effects, but lower doses can also cause results that are significantly different from control [8,9,28]. Consideration and inclusion of graded effects into the mapping is an appropriate next step for this type of analysis.

This analysis is also restricted to single applications of a PBMT dose at specific wavelength bands. In many studies, multiple doses of PBMT are provided to achieve pain relief as well as irradiation paradigms at different wavelengths (although 633 nm and 808–830 nm are the common wavelengths for analgesia and pain relief with PBMT [23,24]). Future work could incorporate additional dimensions, such as data from multiple-dose experiments (number of applications, duration between applications, etc.) or other wavelengths.

This modeling study developed PhoTOS maps based on experiments that used PBMT to reduce pain or inhibit action potentials. Additional PhoTOS maps could be generated for various other PBMT effects, and candidates include: wound healing, blood flow, neurodegeneration, anti-inflammatory effects, and improved recovery from neural injury.

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The current analysis was restricted to continuous wave application of light for simplicity and necessity (only 1 study using pulsatile light delivery met other criteria). Eventually, incorporating pulsatile schemes would be useful since existing implantable pulse generators could be used to power light delivery devices with pulses [15], but do not have continuous wave capability.

5. Conclusion

To understand effective PBM dosing, it is necessary to know both the amount and wavelength of the light irradiating the target tissue and the desired target effect. Using photon propagation software to estimate the irradiances at the target tissue and selecting PBM studies that met specific criteria to reduce sources of variability, we mapped dose properties and effects observed using an analysis technique we called PhoTOS. Clusters of effective dosing parameters became more discernible than when analyzing responses based on emitter parameters alone. The method was used to successfully choose parameters for a pre-clinical study and the modeling approach was validated through a comparison with an excised tissue model. The PhoTOS analysis was then used to provide approximate power and time requirements to achieve the selective neural inhibition effect. Future experiments would help fill gaps in the PhoTOS map for pain treatment and could include other useful dosing dimensions.

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Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests.

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Data availability

Data will be made available on request.

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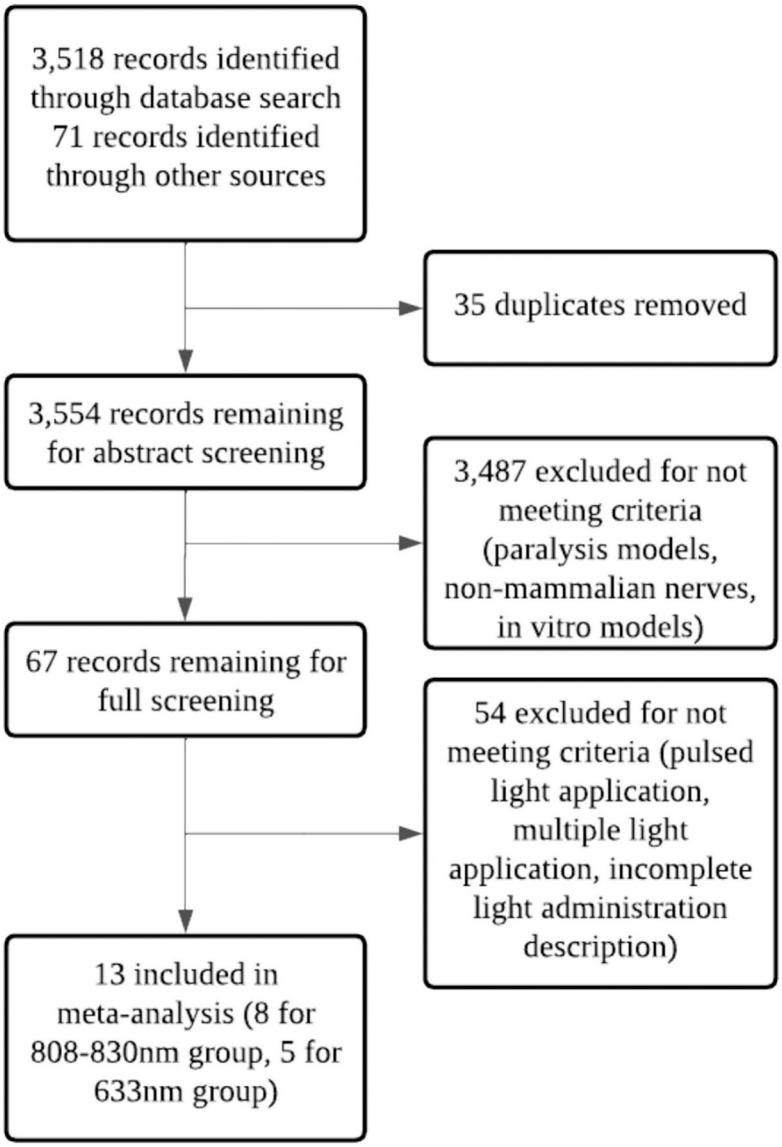


Fig. 1.
Search and selection approach to identify papers for PhoTOS analysis.

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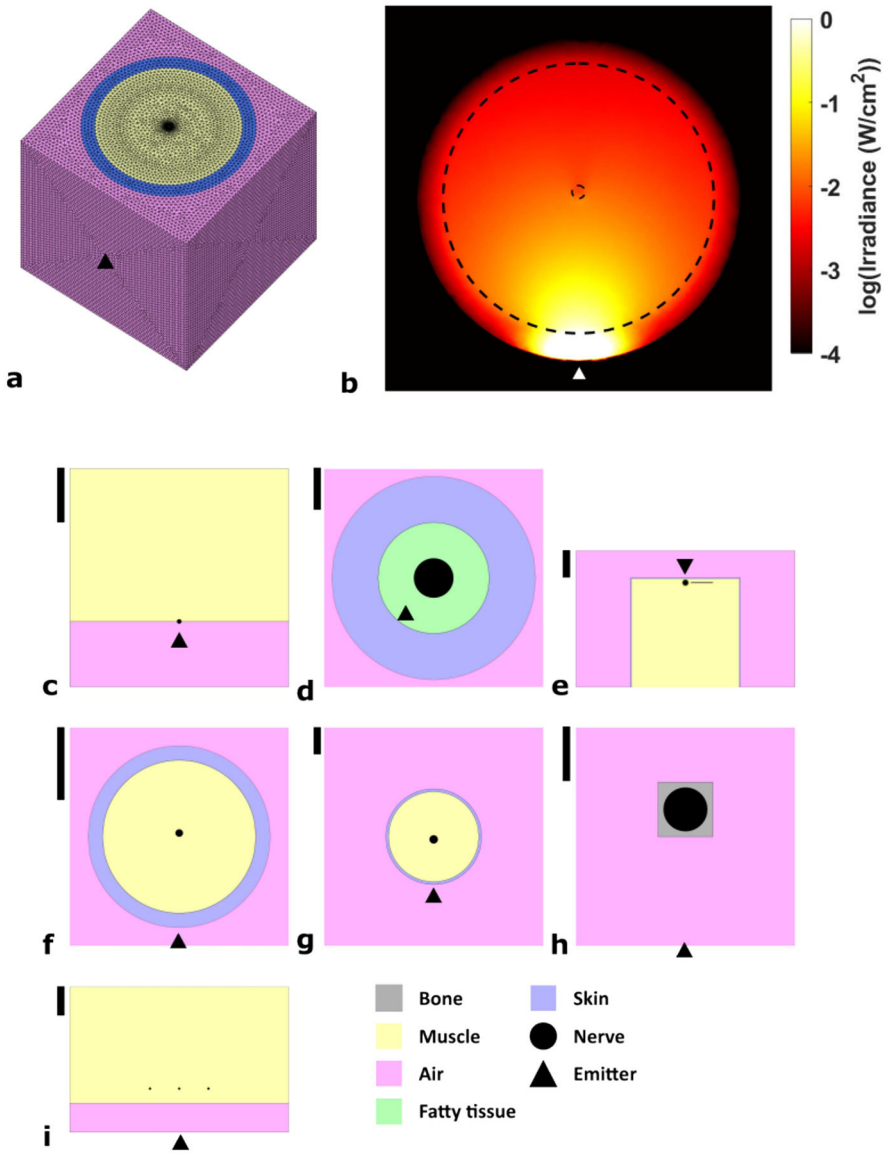


Fig. 2.

MMCLab models used for the PhoTOS method and an example output. (a) Meshed 3D model of a sciatic nerve surrounded by muscle with a layer of skin representing the experiment of Rochkind et al., 1989. (b) False color map of the irradiance simulated with the photon propagation model from (a). (c-h) Experimental procedures (808–830 nm, see Table 1 for experimental procedure identifiers) modeled in COMSOL. (c) Experimental procedures 1, 2, and 3 (irradiation on exposed sural nerve with muscle underneath, scale bar 5 mm). (d) Experimental procedure 4 (irradiation in the intraforamenal space on a dorsal root ganglion, scale bar 10 mm). e) Experimental procedure 5 (irradiation on the lumbar section of the back with one spinal nerve extending beyond the bone, scale bar 10 mm), only one dorsal root from the spinal cord was modeled due to computational memory constraints. (f) Experimental procedure 6 (irradiation on the leg with the sciatic nerve surrounded by muscle and skin, scale bar 10 mm). (g) Experimental procedure 7

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(irradiation on the forearm with the median nerve being surrounded by muscle and skin, scale bar 5 mm). (h) Experimental procedure 8 (irradiation of the tooth with the pulp being surrounded by dentin, scale bar 5 mm). (i) Experimental procedure 9 (irradiation of the paw with the cutaneous nerves being surrounded by muscle under skin, scale bar 2 mm). Black-colored areas indicate nerve, grey-colored areas indicate bone, magenta-colored areas indicate air, yellow-colored areas indicate muscle tissue, blue-colored areas indicate skin, and green-colored areas indicate fatty tissue. All emitters are located at the bottom of each figure pointed to the top of the figure except for (d), in which the emitter is located between the nerve and the bone, modeled after electrode placement in the bony foramen surrounding the dorsal root ganglion, see ref. 44 for details.

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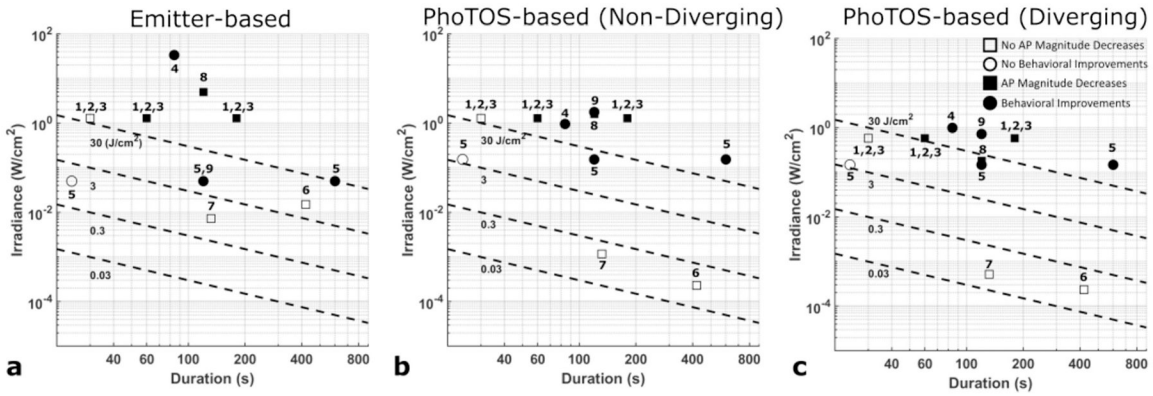


Fig. 3.

Maps plotting effective and ineffective experimental results (810–830 nm) on axes of irradiance and PBM application duration. (a) Data from the standard approach plotted by emitter irradiance. (b) and (c) are PhoTOS maps, because they plot the estimated irradiance at the target nerve, using models with non-divergent (b) or divergent (angle of 22.95°) (c) light, respectively. Note that the PhoTOS maps show a much clearer separation of effective and ineffective experimental data. For a, b, and c, filled symbols represent data that are statistically significantly different from control values and unfilled symbols indicate no significant difference from control. Circles reflect behavioral evaluation, and squares reflect changes in action potential magnitude. Dashed lines show constant photon energy density. Numbers near the markers are associated with experimental procedure identifiers in Table 1.

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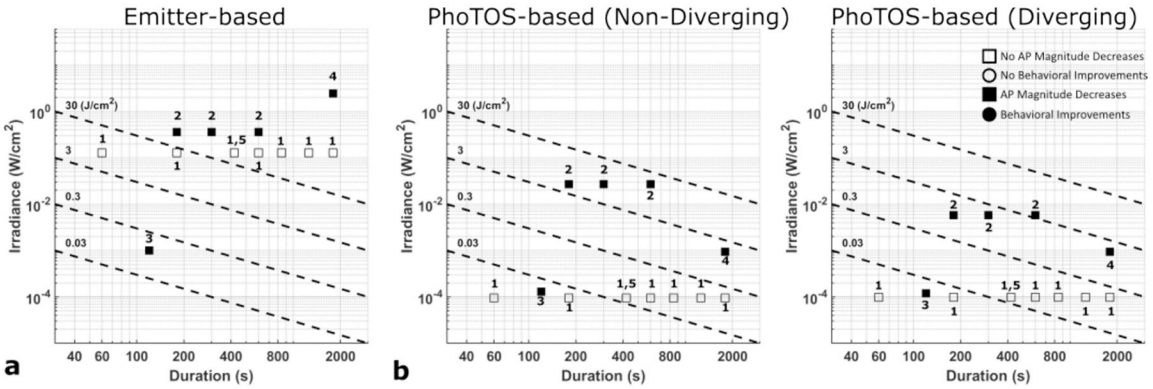


Fig. 4.

Maps plotting effective and ineffective experimental results (633 nm) on axes of irradiance and PBM application duration. (a) Data from the standard approach plotted by emitter irradiance. (b) and (c) are PhoTOS maps, because they plot the estimated irradiance at the target nerve, using models with non-divergent (b) or divergent (angle of 22.95°) (c) light, respectively. Note that the PhoTOS maps show a much clearer separation of effective and ineffective experimental data.For a, b, and c, filled symbols represent data that are statistically significantly different from control values and unfilled symbols indicate no significant difference from control. Circles reflect behavioral evaluation, and squares reflect changes in action potential magnitude. Dashed lines show constant photon energy density.Numbers near the markers are associated with experimental procedure identifiers in Table 2.

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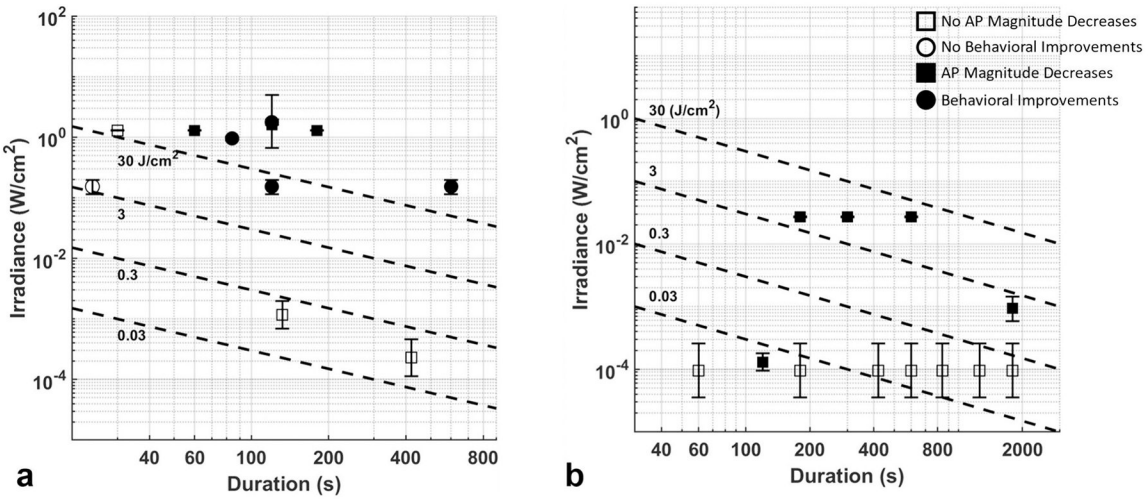


Fig. 5.

PhoTOS sensitivity maps based on the approach shown previously (Fig. 3b and 4b) but calculated using absorption and scattering coefficients that were increased or decreased by +/− 20%. Error bars reflect the maximum (decreased coefficient values) and minimum (increased coefficient values) ends of the range for each data point, reflecting the (a) sensitivity of PhoTOS for 810–830 nm experimental procedures, and the (b) sensitivity of PhoTOS for 633 nm experimental procedures. Note that the sensitivity maps did not change the general structure of the maps, but the potential variability associated with some experimental paradigms. The low end of the irradiance ranges was relatively high.

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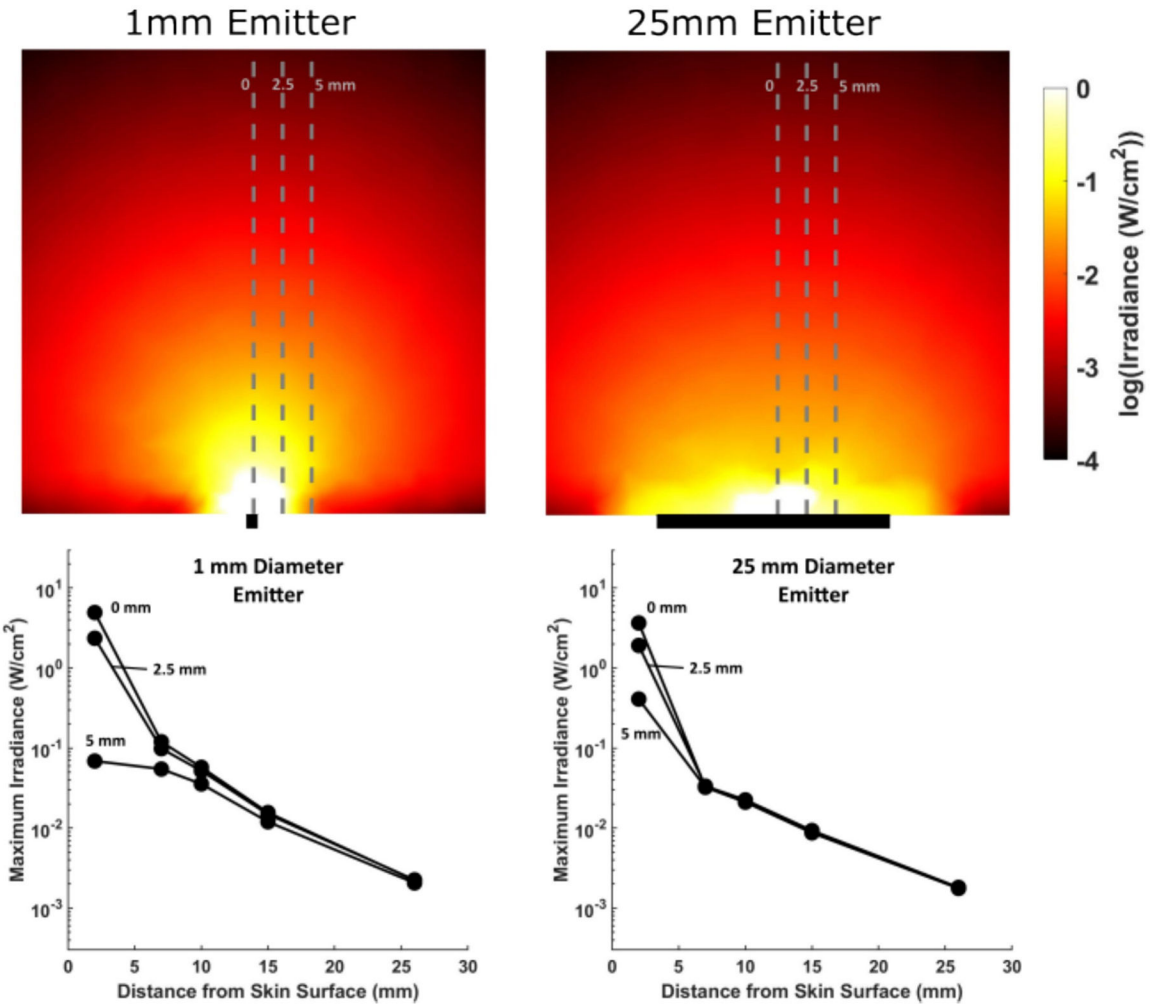


Fig. 6.

Irradiance values at varying tissue depths predicted by a computational model with 1 W emitted from an emitter with a diameter of 1 mm (left) or 25 mm (right). (Upper) Energy flux false color map based on log scale with the emitter sizes shown as the black bars beneath the figures. (Lower) Energy flux as a function of depth from the skin surface along trajectories that are offset 0, 2.5, or 5 mm from the emitter axis.

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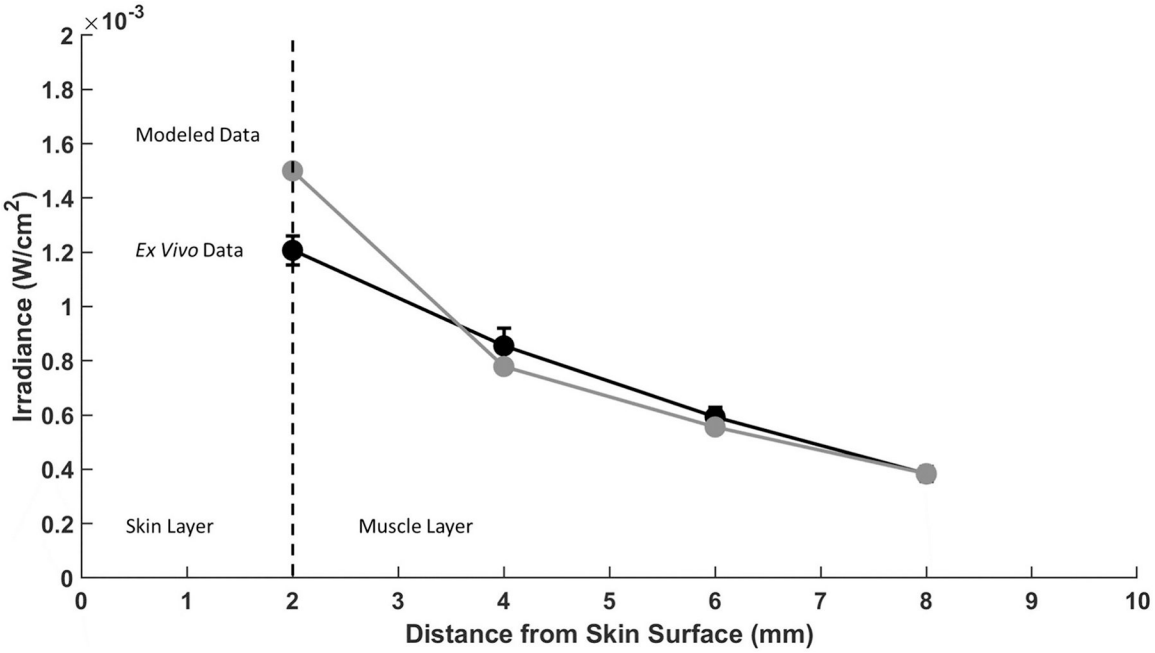


Fig. 7.

Comparison of ex vivo average irradiance measurements (emitter diameter of 1 mm, power of 1 W) with a computational model of the same scenario. ● shows the model irradiance values; ● shows the ex vivo tissue irradiance calculated from measurements ($n = 4$). The optical properties used were the same as in Table 3. Error bars depict +/− SEM.

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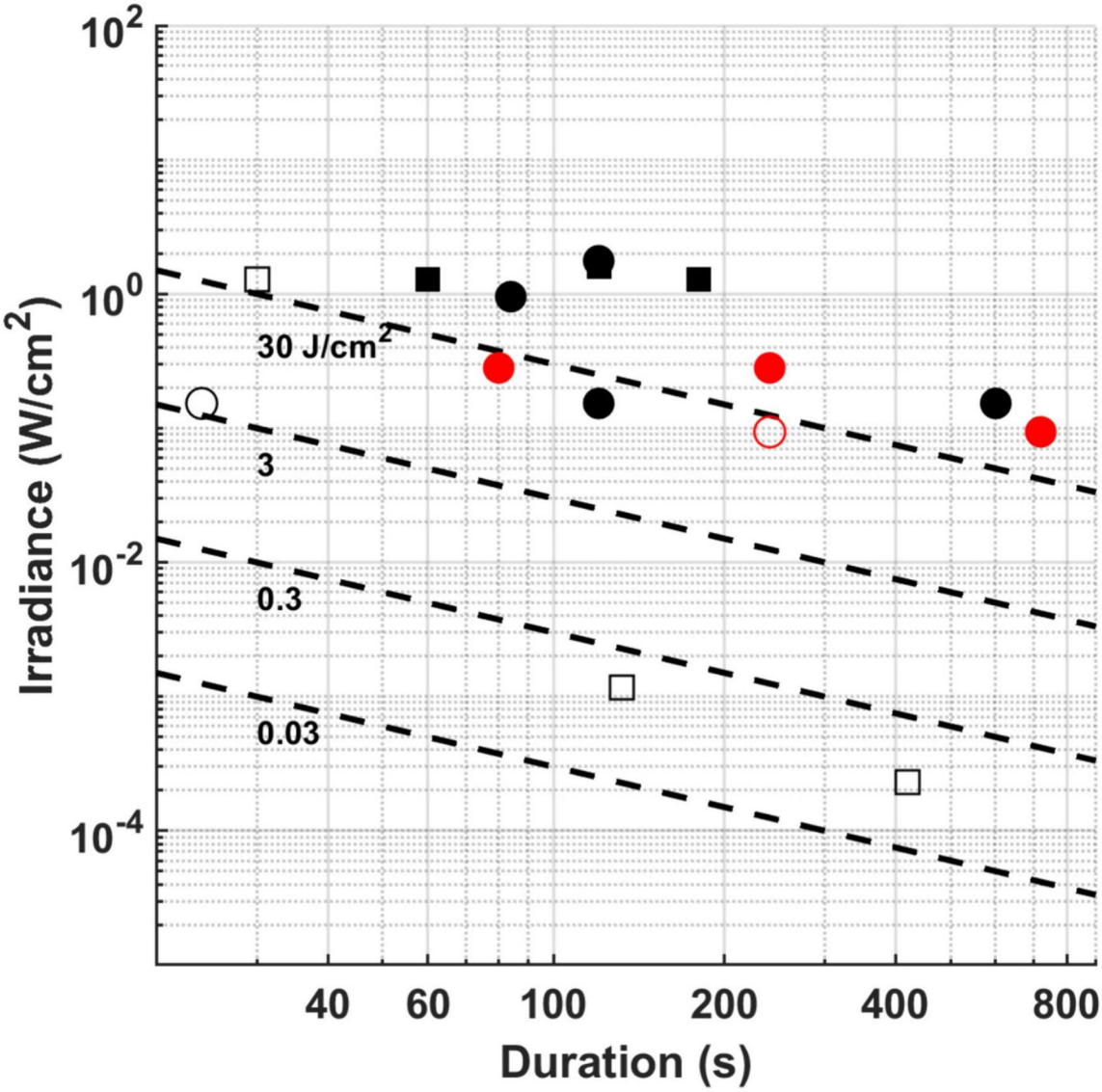


Fig. 8.

Outcomes from a recent study by our group (red) (see ref. 11) on a PhoTOS map (800–830 nm; similar to Fig. 3B). Doses from a previously published study [11] was determined through the PhoTOS map, demonstrating the utility of a PhoTOS map. Open circles represent ineffective behavioral outcomes; closed circles represent effective behavioral outcomes; open squares indicate ineffective action potential reductions; closed squares indicate effective action potential reductions.

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Table 1

Included Works with Wavelengths between 808 and 830 nm.

	Author	Year	Wavelength (nm)	Power (mW)	Duration (s)	Neural Target
1	Tsuchiya, et al. [9]	1993	830	40	30–180	Saphenous
2	Tsuchiya, et al. [8]	1994	830	40	180	Saphenous
3	Sato, et al. [10]	1994	830	40	30–180	Saphenous
4	Holanda, et al. [26]	2016	808	100	84	Dorsal Root Ganglion
5	de Souza, et al. [28]	2019	810	300	24–600	Dorsal Root Ganglion
6	Rochkind, et al. [33]	1986	830	15	420	Sciatic
7	Bartlett, et al. [39]	2002	830	90	132	Median
8	Wakabayashi, et al. [30]	1993	830	350	120	Trigeminal
9	de Souza, et al. [28]	2019	810	300	120	Cutaneous Paw Nerves

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Table 2

Included Works with Wavelengths of 633 nm.

	Author	Year	Wavelength (nm)	Power (mW)	Duration (s)	Neural Target
1	Nissan, et al. [40]	1986	632.8	16	60–1800	Sciatic
2	Shimoyama, et al. [31]	1992	632.8	5.5	180–600	Superior Cervical Sympathetic Ganglion
3	Snyder-Mackler and Bork [41]	1988	632.8	1	120	Radial
4	Shimoyama, et al. [32]	1992	632.8	8.5	1800	Dorsal Horn
5	Rochkind, et al. [33]	1989	633	15	420	Sciatic

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Table 3

Optical Coefficients used in MMCLab simulations at 808 to 830 nm.

Tissue	Absorption Coefficient (mm^{-1})	Scattering Coefficient (mm^{-1})	Anisotropy	Index of Refraction
Bone	0.4 [45]	2.5 [45]	0.9 [46]	1.52 [47]
Intraforamenal Tissue	0.1 [48]	21 [48]	0.9 [48]	1.47 [47]
Muscle	0.035 [49]	1.67 [49]	0.94 [49]	1.4 [47]
Skin	0.4 [49]	35 [50]	0.96 [50]	1.42 [47]
Nerve	0.1 [48]	15 [48]	0.9 [48]	1.4 [48]
Air	0.0	0.0	1.0	1.0

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Table 4

Size Parameters used in PhoTOS Models.

Parameter	Value	Reference
Mouse vertebrae diam	1.8 mm	Shinohara, 1999 [51]
Mouse vertebrae thickness	0.05 mm	Akhter et al., 2004 [52]
Mouse spinal cord radius	0.7 mm	Plemel et al., 2008 [53]
Dorsal Root Ganglion diameter, human	5.9 mm	Hasegawa, et al., 1996 [54]
Dura Thickness, human	0.15 mm	Reina, et al., 2007 [55]
Sciatic diameter, rat	1 mm	Restaino, et al., 2014 [56]
Saphenous diameter, rat	0.4 mm	Campos, et al., 2008 [57]
Sciatic diameter, mouse	0.3 mm	Bala, et al., 2014 [58]
Median diameter, human	2.7 mm	Sugimoto, et al., 2013 [59]
Skin Thickness, rat	2 mm	Ye, et al., 2016 [60]
Skin Thickness, mouse	0.45 mm	Azzi, et al., 2005 [61]
Skin to saphenous distance, rat	6 mm	Walzcak, et al., 2005 [62]
Skin to sciatic distance, rat	10 mm	Savastano, et al., 2014 [63]
Skin to sciatic distance, mouse	5 mm	Savastano, et al., 2014 [63]
Skin to medial distance, human	10 mm	McCahon and Bedforth, 2007 [64]
Skin to spinal cord, mouse	1 mm	Householder et al., 2019 [65]
Rat dentin thickness	1.2 mm	Pitaru and Zajicek, 1981 [66]
Rat pulp diameter	0.6 mm	Pitaru and Zajicek, 1981 [66]
Rat paw thickness	10 mm	Kim et al., 2020 [67]
Rat paw cutaneous nerve diameter	0.15 mm	Kovačič et al., 1999 [68]

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Table 5

Estimated powers that are expected to achieve an effective transcutaneous PBM dose based on PhoTOS estimates generated from two doses of Ref. 11.

Depth (mm)	Time (min): 1.33		Time (min): 12	
	Emitter Diameter (mm)		Emitter Diameter (mm)	
	1	25	1	25
2	56.8 mW	76.9 mW	18.9 mW	25.6 mW
7	2.35 W	8.67 W	781.7 mW	2.89 W
10	4.89 W	12.69 W	1.63 W	4.23 W
15	17.98 W	30.29 W	5.99 W	10.10 W